

IDENTIFICATIONCODE : GI-4-S1-EC-SDF
ECTS : 2**HOURS**Cours : 0h
TD : 16h
TP : 12h
Projet : 0h
Evaluation : 0h
Face à face pédagogique : 28h
Travail personnel : 12h
Total : 40h**ASSESSMENT METHOD****TEACHING AIDS****TEACHING LANGUAGE**

French

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This course belongs to teaching unit Option (GI-4-S1-UE-OPTN) and contributes to the following skills :

Knowledge :

- Defining a rationale for dimensioning
- MTBF MTTF MTTR
- State graph, Markov chain
- Fault Tree, Reliability Diagram
- Decision support

Capacities :

- Apprehend the phenomena of malfunctioning
- Preliminary Risk Analysis
- Lifetime, Reliability, Availability, Maintainability of material and organisational structures
- Contract, commitment and penalties
- Negotiate solutions to increase fault tolerance
- Performance indicators
- Dimensioning

CONTENT

Dependability assesment methods

BIBLIOGRAPHYSûreté de Fonctionnement des systèmes matériels, A. Villemeur, Edt Eyrolles
Reliability Eng., Elsevier**PRE-REQUISITES**Stochastic process
Markov Chains
Probability